

# MetroFlow Held-Out Morphology

## External Adversarial Audit and Complete Failure History

Frozen evidence report / 2026-08-23

|                           |                                          |
|---------------------------|------------------------------------------|
| PRIMARY RESULT            | FAIL - 15/18                             |
| CLAIM LEVEL               | Seed-held-out structural morphology only |
| EMPIRICAL CITY VALIDATION | NOT EVALUATED                            |
| TRAFFIC FUNCTION          | NOT COMPLETED / NEGATIVE RESULT          |
| RUNTIME DEFAULT           | PROMOTION BLOCKED                        |

This report preserves the failed grid\_core result, all six generated map samples, the complete MF-001 through MF-043 history, and the distinction between implementation evidence and scientific promotion.

Source commit: e1979df281ac25a24a82fdea725422310f7c6807  
Source tree: 622dbe37790c5414141cd5d5531a2158aa56f7e5  
Scientific fingerprint: 84b2f3657b21c9cedd9fe7723b4edf99ac3f7514cf7128aff6c994ff9da253f1

## 1. Executive decision

The fixed matrix contains six styles and three unseen seeds. Exactly 15 of 18 cases passed, zero were skipped, and grid\_core failed its claim-specific axis-alignment check on seeds 503, 701, and 907. The aggregate protocol verdict is FAIL.

Five non-grid styles passed direct structural falsifiers plus nonempty output, exact style identity, and deterministic replay. Their success supports bounded structural differentiation only. It does not cancel the grid failure or establish empirical realism.

This is NOT fresh-OSM empirical validation. Traffic-functional validation was not completed. Traffic algorithms remain research-incomplete, and runtime-default promotion remains blocked.

PR #17 remains a valid structural implementation/evidence-identity milestone: exact-head push and synthetic-merge CI both passed on tree 622dbe3779. Those checks are software evidence, not a scientific promotion receipt.

MetroFlow is single-developer personal research code. Adversarial means skeptical scientific and code review, not a hostile-operator threat model. Hashes and manifests are reproducibility and accidental-corruption receipts; this package makes no security or tamper-resistance claim.

## 2. Frozen protocol and harness boundary

Included evidence is the six seed-503 map pairs, the complete 18-case result, frozen validator, portable reproducer, source/sample/harness/remote receipts, claim ledger, and 43-record failure inventory.

The supplied coding harness was applied to finite acceptance, reproduction, and independent verification. The supplied research harness was applied to evidence-before-narrative, claim audit, decision gating, and negative-result preservation. The archives were not installed and did not replace the user's request or repository instructions.

No algorithm, threshold, seed split, baseline, or map was changed. No traffic experiment was rerun. No fresh OSM data was acquired. No push, PR, merge, or publication was performed by this report work unit.

## 3. Result matrix

| Style                                              | Passed | Seeds         | Verdict |
|----------------------------------------------------|--------|---------------|---------|
| Grid Core                                          | 0/3    | 503, 701, 907 | FAIL    |
| Ring-Radial                                        | 3/3    | 503, 701, 907 | PASS    |
| River-Constrained                                  | 3/3    | 503, 701, 907 | PASS    |
| Polycentric TOD                                    | 3/3    | 503, 701, 907 | PASS    |
| Superblock Mixed                                   | 3/3    | 503, 701, 907 | PASS    |
| Curvilinear Warped Grid (organic compatibility ID) | 3/3    | 503, 701, 907 | PASS    |

All generic replay and identity checks passed. The three direct grid failures therefore remain dispositive for the aggregate verdict.

## 4. Style-level adversarial findings

### Grid Core

All generic checks passed: networks were nonempty, style identity was bound, one centre was present, two semantic carrier axes existed, and deterministic replay held.

The claim-specific axis\_aligned\_grid\_core check failed on every seed. Observed aligned shares were 54.436%, 54.751%, 54.560%.

The raw witnesses are internally surprising: semantic horizontal and vertical carrier counts sum to the carrier total, yet the independent axis-aligned count is much smaller. This may expose warped geometry, a semantic-to-geometric mismatch, or an over-strict classifier. The current evidence does not choose among them.

### Ring-Radial

Every seed contains multiple closed degree-2 orbital cycles with winding magnitude one and distinct mean radii.

Orbital counts are 9, 12, 11; maximum radial CV across the matrix is 1.059e-06.

This falsifies the former central-cross masquerade, but does not calibrate ring spacing or traffic resilience against real cities.

## River-Constrained

Each held-out network has two-sided centres, three bridge carriers, and three bridge failure groups.

The result establishes a deterministic structural bottleneck grammar. It does not establish evacuation performance or empirical bridge density.

## Polycentric TOD

Every case has three non-collinear centres, three nonempty geometric catchments, and all three expected centre-pair hierarchy paths.

The evidence supports geometric polycentricity. It does not measure transit service, independent demand basins, or functional accessibility concentration.

## Superblock Mixed

The three cases contain 6, 6, and 5 hierarchy-perimeter macrocells. The smallest qualifying minor axis is 968,732 mm and the largest reported aspect ratio is 1.260, defeating the former one-axis strip false positive.

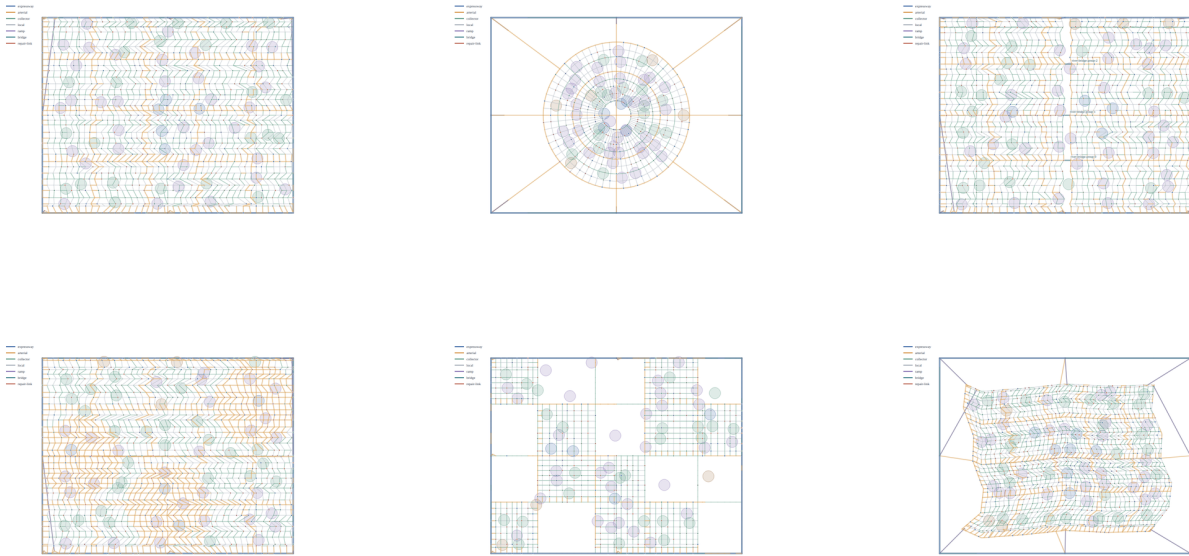
Mode-specific internal access, circulation, walkability, and travel time remain unmeasured.

## Curvilinear Warped Grid (organic compatibility ID)

Bent-connector shares are 98.019%, 98.334%, 97.475%, all above the preregistered 90% threshold.

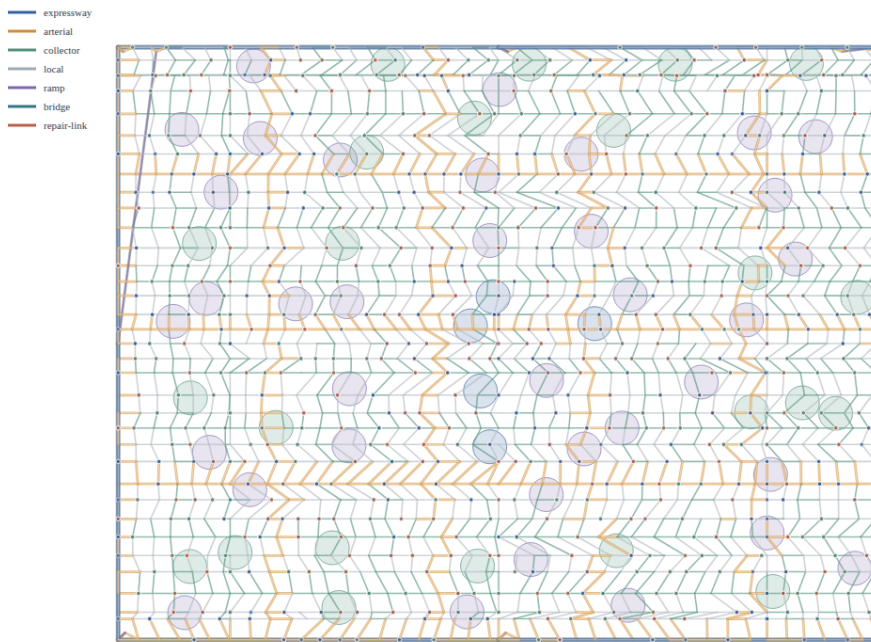
This validates a curvilinear warped-grid embedding only. The report does not claim organic topology, branching history, or empirical urban realism.

## 5. Six-map overview



*Seed 503 contact sheet. Top: grid, ring-radial, river-constrained. Bottom: polycentric, superblock, curvilinear warped grid. Visuals are navigation, not scientific acceptance.*

## Map 1. Grid Core / seed 503



Preview `map_grid_core_s503.png`; SVG authority `map_grid_core_s503.svg`. Nodes 1,652; roads 3,199; case verdict *FAIL*.

### Measured interpretation

All generic checks passed: networks were nonempty, style identity was bound, one centre was present, two semantic carrier axes existed, and deterministic replay held.

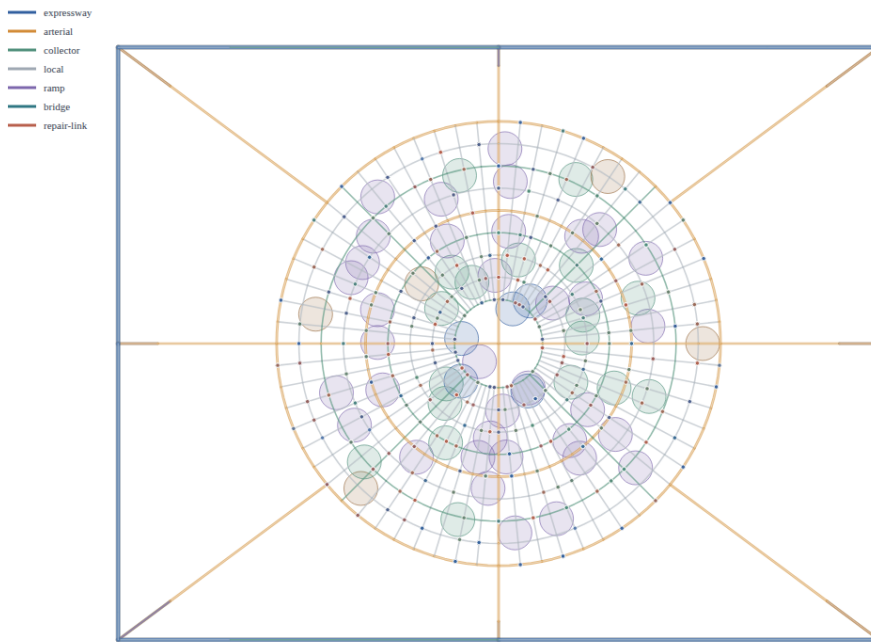
The claim-specific `axis_aligned_grid_core` check failed on every seed. Observed aligned shares were 54.436%, 54.751%, 54.560%.

The raw witnesses are internally surprising: semantic horizontal and vertical carrier counts sum to the carrier total, yet the independent axis-aligned count is much smaller. This may expose warped geometry, a semantic-to-geometric mismatch, or an over-strict classifier. The current evidence does not choose among them.

### Seed-503 network fingerprint

20c142d7bb2571be70dfa4bd67682d8969ad572d55b06f93285f7e0933ae0fe2

## Map 2. Ring-Radial / seed 503



Preview [map\\_ring\\_radial\\_s503.png](#); SVG authority [map\\_ring\\_radial\\_s503.svg](#). Nodes 609; roads 1,140; case verdict PASS.

### Measured interpretation

Every seed contains multiple closed degree-2 orbital cycles with winding magnitude one and distinct mean radii.

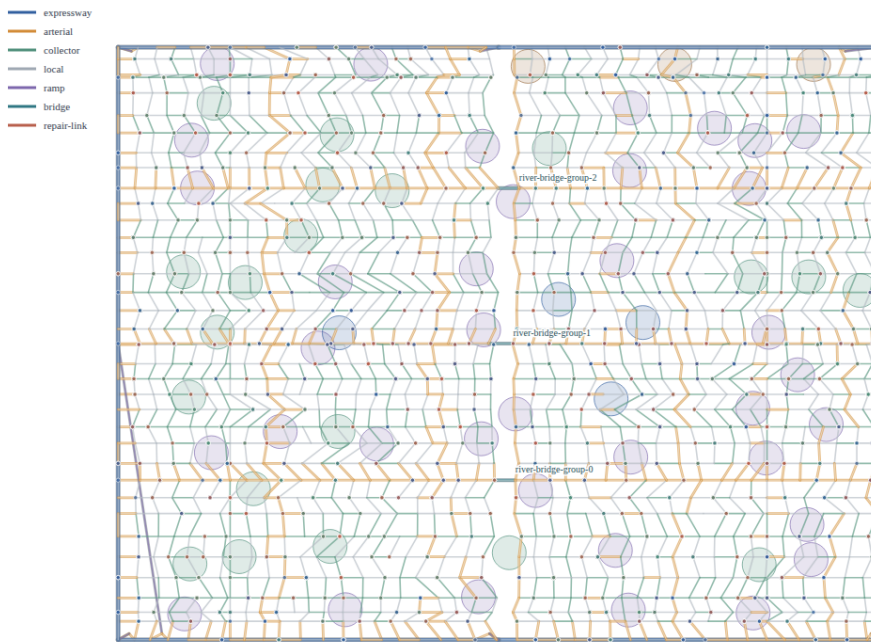
Orbital counts are 9, 12, 11; maximum radial CV across the matrix is 1.059e-06.

This falsifies the former central-cross masquerade, but does not calibrate ring spacing or traffic resilience against real cities.

### Seed-503 network fingerprint

3ac962664b58de3e4b87314952fccbcc77b71a8ec6d07e82ac1d5ef3f2e4f38b

## Map 3. River-Constrained / seed 503



Preview `map_river_constrained_s503.png`; SVG authority `map_river_constrained_s503.svg`. Nodes 1,659; roads 3,175; case verdict PASS.

### Measured interpretation

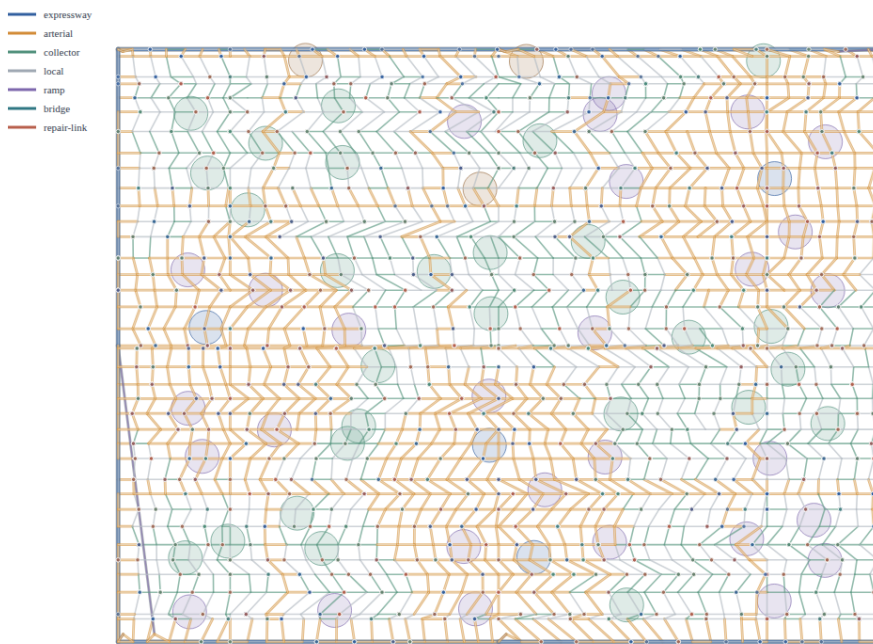
Each held-out network has two-sided centres, three bridge carriers, and three bridge failure groups.

The result establishes a deterministic structural bottleneck grammar. It does not establish evacuation performance or empirical bridge density.

### Seed-503 network fingerprint

fdd0d6a0724bf28e63768c5a4312968dfd92648f95517fda6a5134e35bb04346

## Map 4. Polycentric TOD / seed 503



Preview map\_polycentric\_tod\_s503.png; SVG authority map\_polycentric\_tod\_s503.svg. Nodes 1,747; roads 3,388; case verdict PASS.

### Measured interpretation

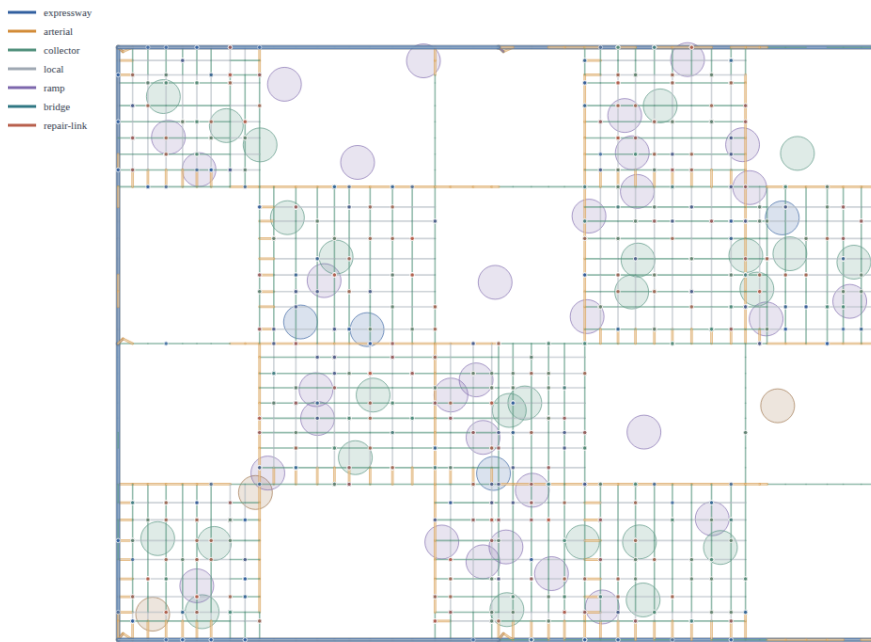
Every case has three non-collinear centres, three nonempty geometric catchments, and all three expected centre-pair hierarchy paths.

The evidence supports geometric polycentricity. It does not measure transit service, independent demand basins, or functional accessibility concentration.

### Seed-503 network fingerprint

b324889185336922b23dccf09ea6d62d6f15af010600a8e99a732c32bda333ea

## Map 5. Superblock Mixed / seed 503



Preview map\_superblock\_mixed\_s503.png; SVG authority map\_superblock\_mixed\_s503.svg. Nodes 1,078; roads 1,913; case verdict PASS.

### Measured interpretation

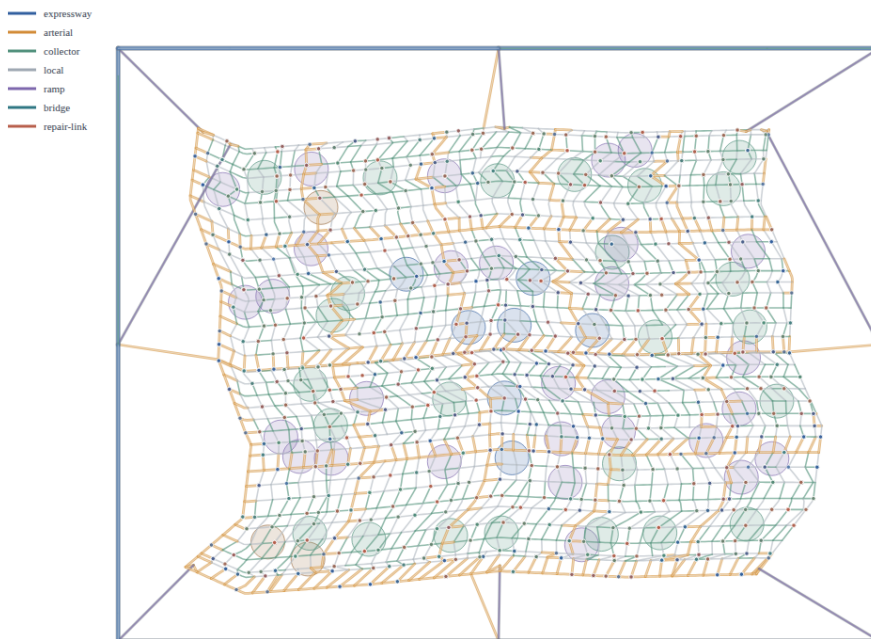
The three cases contain 6, 6, and 5 hierarchy-perimeter macrocells. The smallest qualifying minor axis is 968,732 mm and the largest reported aspect ratio is 1.260, defeating the former one-axis strip false positive.

Mode-specific internal access, circulation, walkability, and travel time remain unmeasured.

### Seed-503 network fingerprint

adf2814f99f37a0881d1faa4dcd6e050102f967c795813f843c802efb3225bfc

## Map 6. Curvilinear Warped Grid (organic compatibility ID) / seed 503



Preview [map\\_organic\\_s503.png](#); SVG authority [map\\_organic\\_s503.svg](#). Nodes 1,675; roads 3,232; case verdict PASS.

### Measured interpretation

Bent-connector shares are 98.019%, 98.334%, 97.475%, all above the preregistered 90% threshold.

This validates a curvilinear warped-grid embedding only. The report does not claim organic topology, branching history, or empirical urban realism.

### Seed-503 network fingerprint

4b06513e5a8aa3d755c67d94a242a4f3104e1149440082167d1c2959f631d6f8

## 6. Claim-evidence decision table

| ID   | Claim                                                                                    | Status          | Boundary                                                                   |
|------|------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------|
| C-01 | The six sample maps and 18 measurements bind to source tree 622dbe3779.                  | VALIDATED       | Identity/provenance only.                                                  |
| C-02 | The held-out inventory contains exactly 18 scored attempts and no skips.                 | VALIDATED       | Completeness of this fixed matrix only.                                    |
| C-03 | All 18 cases reproduced their deterministic network fingerprint.                         | VALIDATED       | Same-code deterministic replay; not historical environment replay.         |
| C-04 | Five non-grid styles satisfy their stated structural checks on all three held-out seeds. | VALIDATED       | Seed-held-out structural morphology only.                                  |
| C-05 | grid_core satisfies its held-out axis-alignment contract.                                | NEGATIVE_RESULT | Failure cause remains contested; threshold is frozen.                      |
| C-06 | The full six-style held-out protocol passes.                                             | NEGATIVE_RESULT | Do not average away the failed style.                                      |
| C-07 | The generated networks reproduce real-city morphology on fresh external cities.          | FORBIDDEN       | NOT fresh-OSM empirical validation.                                        |
| C-08 | The compatibility style implements organic street-network topology.                      | FORBIDDEN       | Use the public warped-grid label.                                          |
| C-09 | Traffic-functional validity was established.                                             | NEGATIVE_RESULT | Traffic-functional validation was not completed.                           |
| C-10 | The traffic algorithms are complete or production-valid.                                 | FORBIDDEN       | Functionality is the next target; completeness is explicitly out of scope. |
| C-11 | scalable_synthetic_v2 is ready to replace the runtime default.                           | FORBIDDEN       | runtime-default promotion remains blocked.                                 |
| C-12 | PR #17 exact-head and synthetic-merge implementation gates passed.                       | VALIDATED       | Remote software evidence; not scientific promotion.                        |
| C-13 | The package establishes security or tamper resistance.                                   | FORBIDDEN       | Hashes identify reproducibility evidence and accidental corruption only.   |

## 7. Complete failure inventory

This appendix mechanically preserves every scoped item, including repaired paths, open debt, unfavorable measurements, unexecuted scientific gates, and contested interpretation.

| State            | Count |
|------------------|-------|
| CLOSED           | 15    |
| PARTIALLY_CLOSED | 2     |
| OPEN             | 15    |
| NEGATIVE_RESULT  | 7     |
| NOT_EVALUATED    | 3     |
| CONTESTED        | 1     |

### MF-001 Legacy gallery used the wrong generator/provenance label

CLOSED

Domain / original severity: evidence identity / P1

Impact: A visually plausible image could be attributed to a generator that did not produce it.

Evidence: Initial external audit supplied by the user; current scalable gallery manifest and renderer provenance checks.

Resolution/current handling: The scalable gallery now declares topology mode, schemas, source fingerprints, counts, and file digests per style.

Remaining gate: Preserve exact provenance checks whenever a gallery schema changes.

Claim effect: Current six-map sample identity is admissible; legacy gallery claims remain historical only.

### MF-002 Gallery accepted score and geometry with a plus/minus ten-node mismatch

CLOSED

Domain / original severity: evidence identity / P1

Impact: Historical scores could be displayed beside a different topology.

Evidence: Initial audit and PR #17 source-topology fingerprint regression tests.

Resolution/current handling: Exact node equality and SHA-256 source-topology fingerprint equality are now required before rendering.

Remaining gate: Retain exact binding in all future renderers and artifact migrations.

Claim effect: The former score-to-shape false-pass path is closed for current artifacts.

### MF-003 CI dependencies and tool versions were not frozen

PARTIALLY\_CLOSED

Domain / original severity: reproducibility / P1

Impact: A frozen candidate could resolve a different Python dependency environment on each run.

Evidence: PR #17 workflow and ci-constraints.txt; exact-head runs 32564391135 and 32564393072.

Resolution/current handling: Ubuntu version, Python 3.12.14, action commit SHAs, and exact Python package versions are pinned.

Remaining gate: Wheel hashes and a runner-image digest remain outside the current closure.

Claim effect: Resolver/toolchain reproducibility is supported, but whole-image bit hermeticity is not.

### MF-004 GitHub-hosted runner and wheels are not bit-hermetic

OPEN

Domain / original severity: reproducibility / P2

Impact: The Ubuntu 24.04 hosted image and downloaded wheels may change beneath version constraints.

Evidence: GitHub Actions workflow uses ubuntu-24.04 and ci-constraints.txt without image or wheel hashes.

Resolution/current handling: No change was made in this report-only work unit.

Remaining gate: Record runner image identity or adopt a digest-pinned container and require-hashes if bit reproduction becomes a release requirement.

Claim effect: CI evidence is exact-head functional evidence, not a bit-hermetic build claim.

**MF-005 Node-to-TAZ assignment allowed float-to-millimetre reconstruction**

CLOSED

Domain / original severity: geometry contract / P2

Impact: Sub-metre coordinates or coercion could change deterministic ownership.

Evidence: PR #17 exact-mm ownership implementation and regression coverage.

Resolution/current handling: Built-in integer x\_mm/y\_mm pairs are now the single authority and squared integer distance is used.

Remaining gate: Keep coordinate-unit validation at public boundaries.

Claim effect: The current Node-to-TAZ internal exact-mm contract is closed.

**MF-006 Project-state documentation was stale**

CLOSED

Domain / original severity: governance / P3

Impact: Reviewers could mistake an older candidate receipt for current state.

Evidence: Updated morphology closure ledgers and PR #17 receipt separation.

Resolution/current handling: The active claim ledger was updated and remote receipts are kept separate from self-referential commits.

Remaining gate: Continue labelling every receipt with its exact commit/tree.

Claim effect: Current report does not inherit a predecessor receipt without an explicit identity link.

**MF-007 Ring-radial style was a grid with a central cross**

CLOSED

Domain / original severity: morphology grammar / P1

Impact: The label claimed ring-radial structure without closed orbital rings.

Evidence: PR #17 nested orbital implementation and winding/radial-CV tests; held-out cases 503, 701, 907.

Resolution/current handling: The generator now emits closed concentric cycles, radial spokes, and a peripheral mainline.

Remaining gate: Calibrate ring segmentation and compare against independent empirical distributions before realism claims.

Claim effect: Structural ring-radial differentiation is validated; empirical realism is not.

**MF-008 Polycentric centres were collinear**

CLOSED

Domain / original severity: morphology grammar / P1

Impact: A one-dimensional chain was presented as a two-dimensional polycentric layout.

Evidence: PR #17 non-collinearity, catchment, and hierarchy-path tests; held-out cases 503, 701, 907.

Resolution/current handling: Three non-collinear centres, nonempty geometric catchments, and centre-pair hierarchy paths are now required.

Remaining gate: Validate independent activity basins and transit/service concentration separately.

Claim effect: Geometric polycentricity is validated, not functional TOD behaviour.

**MF-009 Polycentric functional independence and transit concentration are untested**

NOT\_EVALUATED

Domain / original severity: traffic-function science / P1 scientific

Impact: Geometric centres can exist without distinct accessibility basins or TOD-like transport function.

Evidence: No completed functional-centre or transit-service experiment is present in this evidence package.

Resolution/current handling: No claim was promoted; the limitation is explicit.

Remaining gate: Run preregistered centre-level demand, accessibility, transit concentration, and route substitution tests.

Claim effect: Functional polycentricity remains NOT\_EVALUATED.

**MF-010 Superblock implementation produced one-axis elongated strips**

CLOSED

Domain / original severity: morphology grammar / P1

Impact: A large-area outlier could pass while remaining a thin strip rather than a two-dimensional macroblock.

Evidence: PR #17 two-axis removal, aspect-ratio, minimum-axis, count, and hierarchy-perimeter tests; held-out cases.

Resolution/current handling: Current grammar requires multiple bounded two-dimensional macrocells with hierarchy-qualified perimeters.

Remaining gate: Retain the two-axis structural tests and evaluate internal access separately.

Claim effect: The structural macroblock claim is closed within the current deterministic grammar.

**MF-011 Superblock access, circulation, and mode function are unvalidated**

NOT\_EVALUATED

Domain / original severity: traffic-function science / P1 scientific

Impact: Removing motorized carriers can create a geometric macroblock without usable internal circulation or access.

Evidence: No completed motorized-access, pedestrian-layer, travel-time, or circulation experiment is in scope.

Resolution/current handling: No functional superblock claim was made.

Remaining gate: Define typed internal access and test mode-specific reachability and performance.

Claim effect: Functional superblock validity remains NOT\_EVALUATED.

**MF-012 Organic topology claim exceeded a warped-grid implementation**

CLOSED

Domain / original severity: claim accuracy / P1

Impact: Smooth displacement changed embedding but not branching, dead ends, degree structure, or accretion grammar.

Evidence: Current public label is 'Curvilinear Warped Grid (organic compatibility ID)' and held-out bent-connector checks.

Resolution/current handling: The claim was narrowed to deterministic curvilinear embedding while retaining the compatibility identifier.

Remaining gate: A future organic-topology claim needs branching, angle, degree, dead-end, and block-shape tests.

Claim effect: Curvilinear geometry is validated; organic street-network topology is not claimed.

**MF-013 Morphology semantic tests used weak proxy statistics**

CLOSED

Domain / original severity: test validity / P1

Impact: Orientation entropy or coordinate inequality could pass networks lacking the named structure.

Evidence: PR #17 structural falsifiers for cycles, winding, catchments, hierarchy paths, macroblocks, and curvilinear connectors.

Resolution/current handling: Proxy-only assertions were replaced or bounded by direct structural witnesses.

Remaining gate: Future morphology claims require claim-specific falsifiers, not reuse of unrelated summary metrics.

Claim effect: The six current structural labels have direct tests at their stated claim level.

**MF-014 The v4 control table does not score scalable\_synthetic\_v2**

OPEN

Domain / original severity: scientific evidence / P1 scientific

Impact: A valid historical/sidecar table cannot establish empirical validity for the newly changed scalable generator.

Evidence: morphology-control-table-v4 arm inventory and its explicit diagnostic-only claim boundary.

Resolution/current handling: No silent inference is made from v4 to scalable\_synthetic\_v2.

Remaining gate: Build a preregistered held-out OSM morphology table whose treatment arm is scalable\_synthetic\_v2.

Claim effect: The present 15/18 seed result is structural only and not empirical control-table validation.

**MF-015 Full v4 artifact and manifest were not rederived in CI**

CLOSED

Domain / original severity: reproducibility / P1

Impact: Selected gallery checks could pass while the 135-score table or manifest fields drifted.

Evidence: PR #17 audit-gallery job in runs 32564391135 and 32564393072.

Resolution/current handling: CI now rederives the full v4 JSON/Markdown and verifies schema, table fingerprint, file hashes, and bytes.

Remaining gate: Keep the full matrix command explicit and exact in CI.

Claim effect: Same-head v4 artifact reproducibility is closed for the PR #17 tree.

**MF-016 Score identity omitted or made optional source/spec fields**

CLOSED

Domain / original severity: scientific identity / P2

Impact: Equal metric values could hide different source topology or measurement definitions.

Evidence: Current v4 score payload and constructor tests.

Resolution/current handling: source\_topology\_fingerprint and measurement\_spec are mandatory for current-schema scores and fingerprinted.

Remaining gate: Reject unknown schemas as tracked separately in MF-022.

Claim effect: Current v4 score identity binds both source topology and measurement specification.

**MF-017 PNG previews are digest-bound but not source-rerendered**

OPEN

Domain / original severity: visual evidence / P2

Impact: A committed PNG can match its recorded digest without proving that a current renderer reproduced it.

Evidence: Scalable gallery checker and README distinguish SVG authority from PNG preview.

Resolution/current handling: Success wording now states SVG rederivation and PNG digest comparison separately.

Remaining gate: Rerender PNGs in a pinned raster stack if preview-byte reproduction becomes an acceptance requirement.

Claim effect: PNG maps in this package are navigational previews; SVG/source fingerprints carry scientific identity.

**MF-018 Exact branch-head external CI was initially unavailable**

CLOSED

Domain / original severity: remote verification / P1

Impact: Local evidence alone could not establish that the proposed merge tree passed protected CI.

Evidence: PR #17 push run 32564391135 and pull-request run 32564393072; both ci gate jobs succeeded.

Resolution/current handling: Exact head 6d6e056 and synthetic merge were tested and shared tree 622dbe3779; PR #17 merged as e1979df.

Remaining gate: Obtain new exact-head evidence for any future source change.

Claim effect: PR #17 remote implementation evidence is closed; it does not validate this later report branch remotely.

**MF-019 Duplicate score keys could inflate summaries and overwrite diagnostics**

CLOSED

Domain / original severity: artifact schema / P1

Impact: Two records with one (arm, case) identity could count twice in summaries but collapse in dictionaries.

Evidence: Current MorphologyControlTable constructor and renderer duplicate-key regression tests.

Resolution/current handling: Both the table and renderer reject duplicate (arm, case) keys before aggregation.

Remaining gate: Retain uniqueness as a schema invariant.

Claim effect: The duplicate-key evidence-normalization path is closed.

**MF-020 Catch-all exceptions could be normalized as skipped cases**

CLOSED

Domain / original severity: artifact collection / P1

Impact: Programming or schema failures could be committed as apparently legitimate missing data.

Evidence: Current typed SkippedCase collection and exact 135 score plus 22 skip partition tests.

Resolution/current handling: Unsupported standard styles are classified before generation; only UnsupportedOSMTagError is converted; all other exceptions abort.

Remaining gate: Keep exact score/skip/attempt partition equality and typed exceptions.

Claim effect: Unexpected program failures can no longer silently become normal skips in this collector.

**MF-021 Validation ledger contradicted live remote verification**

CLOSED

Domain / original severity: governance / P1 governance

Impact: The canonical ledger said no push or PR existed after exact-head CI had succeeded.

Evidence: Updated validation ledger plus PR #17 remote receipt in this package.

Resolution/current handling: The predecessor receipt and later review-fix receipt are explicitly separated by commit identity.

Remaining gate: Keep live external receipts outside self-referential commit content and bind them by SHA/tree.

Claim effect: The PR #17 ledger contradiction is closed.

**MF-022 Unknown control-table schema names bypass current identity guards**

OPEN

Domain / original severity: artifact schema / P2

Impact: A typo schema string can avoid v4 source/spec requirements because enforcement is equality-gated.

Evidence: src/metroflow/city/morphology\_control\_table.py current-schema conditional.

Resolution/current handling: No source change was authorized in this report work unit.

Remaining gate: Reject schemas outside an explicit supported-version set or split legacy readers from the current writer.

Claim effect: Unknown-schema inputs are not covered by the v4 identity claim.

**MF-023 Standard supported-style capability has a duplicated source of truth**

OPEN

Domain / original severity: maintainability / P2

Impact: The benchmark can keep skipping a style after the standard generator gains support.

Evidence: STANDARD\_SUPPORTED\_STYLES in the control-table benchmark versus generator capability.

Resolution/current handling: No source change was authorized in this report work unit.

Remaining gate: Export a generator capability registry or assert equality in a focused invariant test.

Claim effect: The current matrix is correct, but future capability drift remains possible.

**MF-024 SkippedCase reason\_code remains a free string**

OPEN

Domain / original severity: artifact schema / P2

Impact: A new typo or undeclared reason can enter code even though the committed artifact is exact-tested.

Evidence: SkippedCase declaration and current artifact reason distribution.

Resolution/current handling: Current production paths emit exactly three reviewed codes and artifact bytes pin their distribution.

Remaining gate: Use a SkipReason enum or enforce an allowlist at construction.

Claim effect: Current receipts are trustworthy; the type-level future extension boundary remains open.

**MF-025 Validation ledger local test count labels a predecessor head**

OPEN

Domain / original severity: governance / P2

Impact: Readers can misread 1,473 passed as the latest head result rather than the predecessor candidate.

Evidence: VALIDATION\_LEDGER.md predecessor receipt and PR #17 review summary reporting 1,483 passed.

Resolution/current handling: Commit identity makes the older number technically scoped, but the prose has not been relabelled in this work unit.

Remaining gate: Label initial-candidate and review-fix local results on separate lines.

Claim effect: Do not use the 1,473 count as a current-tree full-suite receipt.

**MF-026 Seven-metric envelope admits a morphology null operator**

OPEN

Domain / original severity: scientific validity / P1 scientific

Impact: A sidecar local-fabric control can pass all seven empirical metrics while remaining visibly non-city-like.

Evidence: artifacts/external\_audit\_2\_maps and morphology-control-table-v4 diagnostic results.

Resolution/current handling: The result is disclosed as a falsifier of sufficiency; no realism promotion is made.

Remaining gate: Add independently justified discriminators and held-out controls before empirical morphology claims.

Claim effect: Passing the v4 metric envelope is not sufficient evidence of urban realism.

**MF-027 orientation\_order is vacuous in the current empirical envelope**

OPEN

Domain / original severity: measurement validity / P2 scientific

Impact: Near-universal coverage makes the metric unable to falsify treatment maps.

Evidence: CLAIM\_LEDGER.md and morphology-control-table-v4 diagnostics report 0.998 coverage.

Resolution/current handling: The metric is reported rather than counted as silent passing evidence.

Remaining gate: Recalibrate, replace, or preregister a non-vacuous threshold on independent data.

Claim effect: orientation\_order does not support a morphology-validity claim.

**MF-028 Geometry/topology diagnostics are reported but not gated**

OPEN

Domain / original severity: measurement validity / P2 scientific

Impact: Unregistered geometry touches and other diagnostics can coexist with a nominal table pass.

Evidence: v4 diagnostics, including growth\_fabric\_v1 cases with up to 23 unregistered touches.

Resolution/current handling: Diagnostics are disclosed and not misrepresented as acceptance gates.

Remaining gate: Define justified thresholds and ownership semantics before gating them.

Claim effect: Reported diagnostics cannot be treated as passed scientific criteria.

**MF-029 realistic\_synthetic\_v1 passed zero of 30 morphology cases**

NEGATIVE\_RESULT

Domain / original severity: negative scientific result / P0 promotion blocker

Impact: The proposed realistic generator was too highly connected and had too few dead ends across the fixed matrix.

Evidence: artifacts/runtime\_spine\_review/realistic-city-pr62-plausibility and CLAIM\_LEDGER.md.

Resolution/current handling: The negative result was preserved and default promotion remained blocked.

Remaining gate: A new generator revision needs a preregistered, independent empirical protocol; do not tune against this failed holdout.

Claim effect: The PR62 empirical morphology promotion claim is rejected.

**MF-030 realistic\_synthetic\_v1 failed 100k generation and throughput gates**

NEGATIVE\_RESULT

Domain / original severity: negative runtime result / P0 promotion blocker

Impact: All three seeds missed generation wall, realized population, and paired throughput requirements.

Evidence: PR63 canonical runtime artifacts summarized in CLAIM\_LEDGER.md.

Resolution/current handling: The negative result was preserved; standard remained the runtime default.

Remaining gate: Rework and remeasure using a new versioned protocol rather than weakening the failed gate.

Claim effect: Runtime-default promotion remains blocked.

**MF-031 spacing\_scale 3.0 was fitted to a repaired bug rather than rederived**

OPEN

Domain / original severity: calibration debt / P2 scientific

Impact: A parameter calibrated against defective measurements may encode the defect rather than urban morphology.

Evidence: artifacts/external\_audit\_2\_maps/README.md calibration caveat.

Resolution/current handling: The parameter is disclosed as historical debt, not empirical validation.

Remaining gate: Rederive it under a preregistered measurement specification and independent calibration set.

Claim effect: No scientific inference may rely on spacing\_scale 3.0 as a validated calibration.

**MF-032 The control table lacks a block-size discriminator**

OPEN

Domain / original severity: measurement gap / P2 scientific

Impact: Networks with implausible block scale can satisfy the existing seven metrics.

Evidence: artifacts/external\_audit\_2\_maps/README.md and the null-operator result.

Resolution/current handling: The missing dimension is disclosed; no surrogate metric is silently substituted.

Remaining gate: Add a source-compatible block/face-size statistic with held-out calibration.

Claim effect: The present empirical envelope is incomplete for block morphology.

**MF-033 Held-out grid\_core failed the axis-alignment contract in all three seeds**

NEGATIVE\_RESULT

Domain / original severity: negative held-out result / P1 scientific

Impact: Only 1,724/3,167, 1,740/3,178, and 1,681/3,081 semantic carriers were counted as axis-aligned.

Evidence: evidence/heldout\_morphology\_results.json cases grid\_core seeds 503, 701, and 907.

Resolution/current handling: The aggregate verdict is FAIL at 15/18; thresholds and results were not changed after observation.

Remaining gate: Diagnose semantics versus geometry under a new preregistered protocol before modifying implementation or criterion.

Claim effect: The grid\_core held-out structural claim is a NEGATIVE\_RESULT.

**MF-034 Grid-core failure has unresolved construct-validity ambiguity**

CONTESTED

Domain / original severity: scientific interpretation / P1 scientific

Impact: The failed ratio may indicate genuinely warped grid carriers or an over-strict classifier that excludes intended connectors.

Evidence: heldout validator source plus grid\_core witness counts; no independent block-orientation measure was run.

Resolution/current handling: The failure is preserved without selecting the favorable explanation.

Remaining gate: Predeclare a diagnostic decomposition, inspect source semantics, and use an independent orientation/block witness on new seeds.

Claim effect: Cause is CONTESTED; neither generator defect nor invalid test is established by this result alone.

**MF-035 Held-out validation is seed-only, not fresh OSM or named-city validation**

NOT\_EVALUATED

Domain / original severity: scope limitation / P1 scientific

Impact: Unseen pseudo-random seeds test replay and grammar generalization but not external urban realism.

Evidence: Result protocol declares holdout\_kind seed\_heldout\_structural\_not\_fresh\_osm and fresh\_osm\_holdout\_status NOT\_EVALUATED.

Resolution/current handling: The report names this limitation in every promotion decision.

Remaining gate: Run a leakage-controlled fresh-OSM/named-region comparison with licensed frozen extracts and preregistered metrics.

Claim effect: This is NOT fresh-OSM empirical validation.

**MF-036 Traffic-functional development probe did not reach a valid result**

NEGATIVE\_RESULT

Domain / original severity: negative traffic result / P1 research blocker

Impact: The only durable test cache records failure of the nonvacuous-and-directional probe; no complete matrix or durable result artifact exists.

Evidence: /tmp/urban-traffic-heldout-functional-20260822 pytest lastfailed receipt and session record; no result JSON/Markdown was emitted.

Resolution/current handling: The attempt is preserved as a NEGATIVE\_RESULT and was not rerun, repaired, or normalized in this report work unit.

Remaining gate: Design a new versioned functional protocol with nonvacuity, cache invalidation, conservation, deterministic replay, and baseline comparison before execution.

Claim effect: Traffic-functional validation was not completed; traffic algorithms remain research-incomplete and no completeness claim is permitted.

**MF-037 OSM controls are a leaked development set with a different sampled population**

OPEN

Domain / original severity: empirical-control validity / P1 scientific

Impact: The same extracts informed spacing\_scale calibration, and 5-12 km2 core bounding boxes are not comparable to whole-municipality reference statistics.

Evidence: docs/harness/CLAIM\_LEDGER.md control-population audit; Charlotte, Seoul, and Chicago orientation-order comparisons.

Resolution/current handling: The control arm is disclosed as development data rather than an independent positive control.

Remaining gate: Freeze a leakage-free calibration/holdout split and compare equivalent geographic populations under one measurement specification.

Claim effect: Existing OSM-control agreement cannot support held-out empirical generalization.

**MF-038 The default strict OSM importer rejects six of seven control extracts**

OPEN

Domain / original severity: external-data ingestion / P1 scientific

Impact: A reported five-of-five measurable control result depends on opt-in osm\_wiki policy while the shipped strict default blocks most fixtures.

Evidence: docs/harness/CLAIM\_LEDGER.md records strict-policy failure on 6/7 extracts and osm\_wiki coverage.

Resolution/current handling: The policy dependency is now explicit; unsupported tags fail closed rather than becoming generic skips.

Remaining gate: Define which import policy is scientific authority and validate it on a preregistered external corpus.

Claim effect: OSM-control coverage cannot be attributed to the default importer without qualification.

**MF-039 Historical morphology metrics mixed measurement definitions and depended on representation**

PARTIALLY\_CLOSED

Domain / original severity: measurement validity / P1 scientific

Impact: Endpoint chords, member-edge bearings, polyline splitting, parallel edges, and simplify policy produced different values for equivalent-looking roads.

Evidence: docs/harness/CLAIM\_LEDGER.md measurement-definition audit and v4 OSMnx 2.1.1 parity receipt.

Resolution/current handling: The current v4 names and fingerprints BOEING\_2019\_HO and is parity-tested; historical v1-v3 measurements remain definition-bound records.

Remaining gate: Treat representation choice as part of every metric contract and never compare artifacts across specs without remeasurement.

Claim effect: Current v4 parity is supported, but morphology statistics are not representation-invariant or interchangeable with historical tables.

## MF-040 A historical control-table artifact cannot reproduce at its introducing commit

OPEN

Domain / original severity: historical reproducibility / P1 evidence

Impact: The artifact's introducing commit imports growth\_fabric code that only appears in the following commit and lacks complete provenance.

Evidence: docs/harness/CLAIM\_LEDGER.md identifies morphology-control-table-20260807.json, commit c00c9e9, and later module commit dae8b66.

Resolution/current handling: History was not rewritten; later v4 artifacts carry stronger same-head provenance and rederivation gates.

Remaining gate: Keep the historical artifact labelled unreproducible and do not use it as replay authority.

Claim effect: The historical table is evidence of a recorded result, not same-commit reproducibility.

## MF-041 Legacy growth-fabric river-constrained maps were severed by the river

NEGATIVE\_RESULT

Domain / original severity: negative morphology result / P1 scientific

Impact: The audited maps had zero cross-river links, 53 weak components, and a largest-component share near 0.52 despite bridge/ramp enum names.

Evidence: docs/harness/CLAIM\_LEDGER.md legacy growth\_fabric river-constrained diagnostics.

Resolution/current handling: The historical negative result is preserved; current scalable\_synthetic\_v2 has a separate three-bridge structural witness.

Remaining gate: Do not transfer the current grammar's pass backward to the historical generator; remeasure any future river treatment by version.

Claim effect: Legacy river connectivity claims are rejected even though the current scalable grammar passes its bounded bridge test.

## MF-042 Legacy growth-fabric bypass did not exist as a network function

NEGATIVE\_RESULT

Domain / original severity: negative morphology result / P1 scientific

Impact: Eight tangential expressways formed a forest with cyclomatic number zero; the test passed on a gateway radial rather than a bypass arc.

Evidence: docs/harness/CLAIM\_LEDGER.md bypass topology and angular-coverage diagnostics.

Resolution/current handling: The false semantic test is retained as historical evidence and is not used to support the present six-style result.

Remaining gate: A future bypass claim needs a closed or route-substituting peripheral path with direct topology and traffic witnesses.

Claim effect: No bypass-function or congestion-relief inference is permitted from the historical maps.

## MF-043 The morphology\_quality gate rejected all five of its real controls

NEGATIVE\_RESULT

Domain / original severity: negative gate-validation result / P1 scientific

Impact: Charlotte missed the density floor, all controls failed weak-component unity, four failed block continuity, and the gate had no density ceiling.

Evidence: docs/harness/CLAIM\_LEDGER.md five-control morphology\_quality falsification.

Resolution/current handling: The negative control result is preserved and the gate is not used as promotion authority.

Remaining gate: Redesign the gate against independent real controls with attainable two-sided bounds and representation-compatible measurements.

Claim effect: A generator pass or failure under morphology\_quality cannot establish real-city quality until the gate admits its intended controls.

## 8. Traffic-functional boundary and next protocol

The prior bounded traffic-functional attempt did not produce an admissible result artifact. The only durable witness is a pytest lastfailed entry for the development nonvacuous-and-directional probe. It is recorded as MF-036 NEGATIVE\_RESULT, not silently skipped and not promoted into an algorithm diagnosis.

The user explicitly directs future work toward functionality rather than completeness because the traffic algorithms require additional change and research. A future versioned protocol should test nonvacuity, explicit units, conservation, immutable state transition, deterministic replay, cache invalidation, baseline fallback, directional response, and bounded performance.

Even a passing future functional matrix would not establish universal correctness, calibration, city realism, operational safety, or runtime-default readiness without separate evidence.

## 9. Reproduction and independent challenge surface

The evidence manifest binds every package file except itself. The checker enforces exact recursive membership, source/tree/fingerprint identity, the 18-case Cartesian partition, all 43 failure identifiers and fields, complete PNG structure, PDF-copy identity, link presence, and claim-ceiling phrases.

The byte-frozen original validator retains its historical author paths because its SHA-256 is evidence. A separate portable reproducer accepts explicit checkout/package paths, verifies source commit and tree, and returns success when bytes reproduce even though the scientific verdict is FAIL.

The most valuable independent attack is the grid classifier construct: reproduce the raw geometry without changing the held-out verdict, decompose semantic carrier labels from coordinate orientation, and predeclare an independent block/orientation witness on new seeds. Do not tune the frozen criterion after observing these cases.

A second high-value attack is external morphology: add licensed, frozen, leakage-controlled OSM extracts and metrics that defeat the known null operator. The current v4 table does not score scalable\_synthetic\_v2 and cannot be borrowed as its empirical validation.

## 10. Final gate

| Surface                        | Decision                                                      |
|--------------------------------|---------------------------------------------------------------|
| Report/evidence package        | Admissible only after checker, PDF QA, and independent review |
| Held-out structural morphology | FAIL - 15/18                                                  |
| Fresh-OSM empirical morphology | NOT EVALUATED / BLOCKED                                       |
| Traffic-functional validity    | NOT COMPLETED / NEGATIVE RESULT                               |
| Traffic algorithm completeness | NOT CLAIMED                                                   |
| Runtime-default promotion      | BLOCKED                                                       |